

Database design for Expedition 77 (Tour Guide Booking Platform)

Business Rules

- Every user in the system must have a unique email address.
- A user must register with the required details (name, email, password, phone) before accessing the system.
- Each user has a role which determines their access level in the system (e.g., Admin, Guide, Customer).
- Each user account has a status indicating whether the account is active, inactive, or suspended.
- A user can request a password reset, and each reset request must contain a unique reset token with an expiration time.
- Only users with the Guide role can have a guide profile.
- Each guide can have one guide profile containing experience, bio, languages, expertise, and rating statistics.
- Guides must submit verification documents such as citizenship ID and license number before being approved.
- A guide's verification must be approved by an admin user.
- Each guide can upload multiple portfolio images to showcase their work.
- A guide may offer services for multiple trekking routes.
- Each route must have a name, location, difficulty level, and estimated duration.
- A route can be assigned to multiple guides.
- Each guide can set a custom price per day for each route they offer.
- Guides must specify their availability dates before customers can book them.

- A guide can have multiple availability records, each representing a specific date.
- The availability status indicates whether the guide is available or unavailable on that date.
- Only users with the Customer role can create bookings.
- A customer can create multiple bookings, but each booking belongs to only one customer.
- A guide can receive multiple bookings, but each booking is assigned to only one guide.
- A booking must specify a start date and end date.
- The system calculates the total price based on the guide's price per day and the duration of the booking.
- A booking may include an optional message from the customer to the guide.
- Each booking has a status such as pending, confirmed, completed, or cancelled.
- Each booking may contain multiple booking dates representing the exact days the guide is booked.
- A guide cannot be booked more than once on the same date.
- A customer can leave a review only after completing a booking.
- Each review must be associated with one booking, one guide and one customer.
- A booking can have only one review.
- Reviews contain a rating and optional comment.
- Guide ratings are aggregated to calculate the average rating and total number of reviews.
- Admin users can manage informational pages such as About, Terms, or FAQs.
- Each page stores the admin who last updated it and the update timestamp.
- Visitors can submit contact messages without creating an account.

- Each message must contain the sender's name, email, subject, and message content.

ER Diagram

